

the scene may need a general wash of light, while others may need a spotlight directed at a specific object. 3ds Max provides a wide collection of lighting types to fit all needs.

Spotlights originate from a single point and spread outward in a conical shape. 3ds Max has two types of spotlights: Target Spot and Free Spot. The only difference is that a Target Spot has the target attribute turned on, which creates a target that the user can manipulate to control the direction of the spot. Spotlights can illuminate both diffuse and specular areas of a scene. They also can cast shadows and decay over distance. Hotspot/Beam adjusts the size of



Spot lights

the entire light; Falloff/Field controls the softness of the edge of the light cone. Spotlights can be either circular or rectangular.

Omni lights can be thought of as bare light bulbs in the room. Like a light bulb, an omni light sends its rays in every direction. Similar to spotlights, omni lights affect the specularity of an object and can cast shadows. The best use for an omni light is as an overall scene light.

Directional lights are similar to spotlights in that they have a specific direction, but a directional light does not emanate from a single



An omnilight in a scene

point. Instead, the rays are parallel and simulate distant lights such as the sun. This makes them ideal for outdoor scenes. This light also comes in two varieties, Target Direct and Free Direct, the only difference being the target.



Directional Light

Area lights cast their light from an area, rather than from a point. These are only available only to the Mental Ray renderer. This makes them a good choice for simulating diffuse lighting, bounce lights, or fluorescent tubes. Although area lights can provide realistic effects, calculating complex shadows can eat up a lot of render time.